

Personal Workout App and Coaching System Design

Overview

This document describes a personal workout tracking and coaching app built as a pure local stack using Flutter for the frontend and Python + FastAPI + DuckDB for the backend. The system supports home strength training with Full Body A/B routines, computes weekly analytics, and generates rule-based coaching suggestions, with an optional local LLM coaching layer.

Fitness Context and Goals

The app targets a user training primarily at home who wants to build muscle using bodyweight and simple loading (backpack, bands, dumbbells). Training focuses on three full-body strength sessions per week, optionally four, plus light cardio/mobility days. Evidence-based guidelines suggest that 3 weekly full-body sessions with progressive overload and 4–12 hard sets per major muscle group are effective for strength and hypertrophy.

Weekly Training Structure

Day	Type	Focus
Day 1	Full Body A	Push + squat emphasis
Day 2	Cardio / mobility	Cycling, running, stretching
Day 3	Full Body B	Pull + hinge emphasis
Day 4	Rest or light cardio	Recovery
Day 5	Full Body A variant	Slightly heavier or more sets
Day 6	Optional conditioning	Mobility or light cardio
Day 7	Rest	Full recovery

Warm-up & Cool-down Protocols

Proper warm-up and cool-down are non-negotiable components of each session. Dynamic warm-ups prepare joints and muscles better than static stretching before lifting, while cool-down stretching and breathwork accelerate recovery and reduce DOMS. The app surfaces the correct protocol type based on which workout template and activity are selected.

Protocol Selection by Activity

Activity / Session	Recommended Warm-up Types	Recommended Cool-down Types
Full Body A (push + squat)	Dynamic full-body + Lower-body focus + Glute activation	Lower-body static + Recovery walk + Breathwork
Full Body B (pull + hinge)	Dynamic full-body + Upper-body focus + Posterior chain activation	Upper-body static + PNF hamstring/lat + Breathwork
Running / cardio	Sport-specific running warm-up + Hip mobility	Light walk + Lower-body static + Diaphragmatic reset
Cycling	Sport-specific cycling warm-up + Hip flexor / ankle mobility	Quad/hip flexor static + Recovery walk
Strength deload / mobility day	Foam rolling + Thoracic mobility	Full-body static + Savasana

Type 1 — Dynamic Warm-up

Active, movement-based exercises that raise core temperature and prepare joints before strength work. Dynamic warm-ups are the baseline for every strength session and should always precede static or PNF stretching on training days. A meta-analysis found that including self-myofascial release (foam rolling) followed by a dynamic warm-up improves overall training performance.

Full-body dynamic routine (7–10 min — use before every strength session):

1. March or jog in place — 60 s (raise heart rate gradually)
2. Arm circles — 10 reps each direction (shoulder joint prep)
3. Leg swings front-to-back — 10 reps each leg
4. Bodyweight squat — 10 reps, slow and controlled
5. Hip circles — 10 reps each leg (loosen hip capsule)
6. Inchworm to push-up — 5 reps (whole-chain activation)
7. World's greatest stretch — 3 reps each side (thoracic + hip)

Lower-body add-on (5–6 min — Full Body A only):

1. Walking lunge with rotation — 5 reps each side
2. Lateral lunge — 8 reps each side (adductor warm-up)
3. Glute bridge — 10 reps, pause 1 s at top
4. Bodyweight clamshell — 12 reps each side
5. Heel-to-toe walk — 10 steps (ankle prep)

Upper-body add-on (4–5 min — Full Body B only):

1. Band pull-apart or arm swings — 15 reps
2. Shoulder rotation (internal / external) — 12 reps each

3. Cat-cow — 8 cycles (thoracic mobility)
4. Thread-the-needle — 6 reps each side
5. Scapular push-up — 10 reps (serratus activation)

Type 2 — Foam Rolling (Self-Myofascial Release)

Foam rolling is a self-myofascial release (SMR) technique that applies pressure to soft tissue to reduce tension, improve short-term range of motion, and support post-exercise recovery. A 2024 study in *Scientific Reports* confirmed that foam rolling significantly accelerates post-exertional recovery compared to passive rest, with rolling duration of at least 120 seconds per area showing meaningful blood lactate clearance. A NIH meta-analysis found foam rolling improves muscular performance and flexibility and alleviates muscle fatigue and soreness. Use foam rolling **before** dynamic warm-up to release soft tissue restrictions, or **after** training as part of the cool-down.

Pre-session foam rolling (4–6 min total — roll each area 60–120 s):

- Thoracic spine (mid-back) — arms crossed over chest, roll 3–4 segments
- Quads — prone, roll from hip to just above knee
- IT band / lateral thigh — side-lying, slow passes from hip to just above knee
- Lats — side-lying, arm overhead, roll along the side of the torso

Post-session foam rolling (5–8 min — pair with cool-down stretch):

- Calves — seated or floor, roll from ankle to knee
- Hamstrings — seated on roller, roll from knee to glute
- Glutes / piriformis — seated, cross one ankle over opposite knee, lean toward that side
- Upper back — same as pre-session thoracic roll, 60–90 s

Key technique rules:

- Move slowly — approximately 1 inch per second over the target tissue
- Pause on tender spots for 20–30 s rather than rolling rapidly back and forth
- Avoid rolling directly over joints, the lower back (lumbar spine), or bony prominences
- Grooved and serrated rollers have similar effectiveness to smooth rollers; duration matters more than texture

Type 3 — Mobility / Joint Prep

Targeted joint-preparation routines that address range-of-motion restrictions before loading. These are best inserted between foam rolling and the main dynamic warm-up on days with known tightness.

Hip & ankle mobility (6–8 min — before leg-dominant sessions):

1. 90/90 hip sit — 30 s each side (hip internal rotation)
2. Deep squat hold with chest-up cue — 30–45 s
3. Ankle circles — 10 reps each direction per ankle
4. Pigeon pose (dynamic, not held) — 8 reps each side
5. Lunge with ankle rock — 6 reps each side

Thoracic spine & shoulder (5–7 min — before push / pull sessions):

1. Foam roll mid-back — 60 s (3–4 segments)
2. Thoracic rotation in quadruped — 8 reps each side
3. Doorway or band chest stretch (dynamic) — 10 reps
4. Prone Y-T-W — 8 reps per letter (scapular control)

Type 4 — Muscle Activation

Brief targeted activation patterns that "wake up" key muscles before the main session. Reduces the risk of compensatory movement patterns (e.g., quad dominance during glute exercises).

Glute & posterior chain activation (5 min — before Full Body B):

1. Glute bridge — 2 × 15 reps, squeeze and hold 1 s at top
2. Donkey kick — 12 reps per leg
3. Banded lateral walk — 10 steps each direction (if band available)
4. Single-leg hip hinge (light) — 8 reps each leg

Core & shoulder-girdle activation (4–5 min — before pressing or plank work):

1. Dead bug — 3 reps each side, slow and controlled
2. Plank shoulder taps — 10 reps each side
3. Scapular push-up — 2 × 10 reps
4. Bird-dog — 6 reps each side (anti-extension)

Type 5 — Sport-Specific Warm-ups

When the selected activity is cardio (running, cycling) rather than strength training, the warm-up shifts to activity-specific movement patterns that mirror the biomechanics of the sport.

Running warm-up (8–10 min before a run):

1. Brisk walk — 3–4 min (gradual heart rate elevation)
2. Leg swings front-to-back and lateral — 10 reps each direction per leg
3. High knees — 30 s (hip flexor activation and running pattern prep)
4. Butt kicks — 30 s (hamstring activation)

5. Ankle circles and calf raises — 10 reps each (ankle prep)
6. Hip circles and lateral shuffles — 10 reps each side
7. Easy-pace jog — 3–5 min (gradual ramp to workout pace)

Cycling warm-up (8–10 min before a ride):

1. Easy spinning at very low resistance — 4–5 min (joint lubrication and heart rate ramp)
2. Hip flexor stretch standing (dynamic) — 8 reps each side (counteract hip tightness from seated position)
3. Ankle circles — 10 reps each foot
4. Gentle quad and glute activation — bodyweight squat 2×10 (prime cycling muscles)
5. Moderate-pace effort at 60–70% of target pace — 3–4 min (neuromuscular readiness)

Bodyweight / HIIT session warm-up (6–8 min):

1. Jumping jacks — 30 s
2. Arm circles forward and backward — 10 reps each
3. Squat-to-stand — 8 reps (hip mobility and ankle mobilisation)
4. Lateral shuffle — 10 steps each direction
5. Explosive step touch or light jump — 30 s (prepare for plyometric demand)

Type 6 — Static Cool-down Stretching

Static stretches are most effective post-workout when muscles are warm. Hold each stretch for 30–60 s. Static stretching before training reduces acute force production; it belongs in the cool-down, not the warm-up.

Lower-body static stretching (8–10 min — after Full Body A or running):

1. Standing quad stretch — 30–45 s each leg
2. Kneeling hip flexor stretch — 30–45 s each side
3. Seated hamstring stretch (one leg extended) — 45 s each side
4. Figure-four glute stretch — 45 s each side
5. Wall calf stretch — 30 s each leg (keep back heel flat)

Upper-body & full-body cool-down (8–10 min — after Full Body B):

1. Cross-body shoulder stretch — 30 s each arm
2. Tricep overhead stretch — 30 s each arm
3. Lunge with spinal twist — 30 s each side (lat and thoracic)
4. Child's pose — 45–60 s (back, hip, and lat decompression)
5. Lying torso twist — 45 s each side (spinal mobility)
6. Cat-cow — 6 slow cycles (resets spinal length)

Recovery walk (5–8 min — after any session):

- Slow walk or light cycling to bring heart rate below ~100 bpm before beginning static stretching. Skipping this step increases the risk of lightheadedness after intense sessions.

Type 7 — PNF Stretching (Proprioceptive Neuromuscular Facilitation)

PNF stretching is an advanced flexibility technique originally developed in clinical neuromuscular rehabilitation. It produces greater short-term gains in passive range of motion than static stretching alone by using isometric contractions to trigger autogenic inhibition (the Golgi tendon organ reflex relaxes the target muscle, allowing deeper stretch). PNF is best used in the cool-down on Full Body B days, or on dedicated mobility sessions. It is typically performed with a partner, though self-directed versions exist.

There are three main PNF technique variants:

Hold-Relax (HR)

The most common solo-friendly variant. The target muscle is passively stretched to its end range, held for 10 s, then the user isometrically contracts that muscle against resistance for 6–10 s at 20–30% of maximum force, before relaxing and moving into a deeper passive stretch held for 30 s. Repeat 2–3 cycles.

Example — hamstrings:

1. Lie on back, raise one straight leg until a gentle hamstring stretch is felt, hold 10 s
2. Gently push the leg down against a band or partner's hand (isometric contraction) — 6–10 s, ~25% effort
3. Relax completely, then raise the leg further into the new range of motion
4. Hold the new stretch for 30 s, then repeat from step 1 for 2–3 more cycles

Contract-Relax (CR)

Similar to hold-relax but the contraction is concentric (the limb actually moves through partial range) rather than purely isometric. Produces slightly greater range of motion gains in some studies but requires a partner to resist the movement safely.

Example — hamstrings:

1. Partner raises the leg to end range and holds it (passive stretch, 10 s)
2. User pushes leg toward the floor (concentric hamstring contraction) against partner resistance for 6–10 s
3. User relaxes; partner moves leg into new range of motion
4. Hold new stretch 30 s and repeat 2–3 cycles

Contract-Relax-Antagonist-Contract (CRAC)

The most complex PNF variant. After the contract-relax cycle, the user actively contracts the antagonist muscle (e.g., quadriceps when stretching hamstrings) to drive the limb deeper into the stretch through reciprocal inhibition. Provides the largest acute range-of-motion gains.

Example — hamstrings (CRAC):

1. Partner raises leg to end range, hold 10 s
2. User isometrically contracts hamstring against partner — 6–10 s
3. User immediately contracts quadriceps (pulls leg upward) while partner assists — 6–10 s
4. Find new end range, hold 30 s; repeat 2–3 cycles

PNF cautions:

- Work at 20–30% of maximum force during contractions — higher effort increases injury risk
- Never use PNF immediately before maximal strength work (temporarily reduces force production)
- Best used on cool-down or rest days for mobility development

Type 8 — Breathwork & Nervous System Reset

Breathwork deliberately shifts the nervous system from sympathetic (high arousal, training) to parasympathetic (rest and recovery). Reducing post-training sympathetic tone speeds heart rate recovery, lowers cortisol, and improves sleep quality when performed post-workout.

Diaphragmatic breathing reset (2–3 min — after every session, before stretching):

1. Lie flat on back, hands on belly
2. Inhale through nose for 4 counts — belly rises, chest stays still
3. Exhale slowly through mouth for 6–8 counts — belly falls
4. Repeat 6–10 cycles (approximately 2–3 min total)

The extended exhale is key: the longer-than-inhale breath pattern directly stimulates the vagus nerve and increases heart rate variability (HRV), a marker of parasympathetic activation.

Box breathing (2–4 min — after high-RPE sessions):

1. Inhale through nose — 4 counts
2. Hold breath — 4 counts
3. Exhale through mouth — 4 counts
4. Hold empty — 4 counts
5. Repeat 4–6 cycles

Progressive muscle relaxation / Savasana (3–5 min — after long or demanding sessions):

1. Lie flat, arms at sides, palms up
2. Tense each muscle group for 3 s then fully release: feet → calves → quads → core → hands → shoulders → face
3. Breathe slowly and let every muscle go heavy
4. Stay still for 3–5 minutes

Warm-up / Cool-down Quick Reference

Protocol Type	Phase	Duration	Best Paired With
Dynamic full-body	Warm-up	7–10 min	Every session
Foam rolling (pre)	Warm-up	4–6 min	Before dynamic warm-up
Mobility / joint prep	Warm-up	5–8 min	Before leg or press work
Muscle activation	Warm-up	4–5 min	Before Full Body B
Sport-specific (running)	Warm-up	8–10 min	Before runs
Sport-specific (cycling)	Warm-up	8–10 min	Before rides
Recovery walk	Cool-down	5–8 min	After any session
Lower-body static	Cool-down	8–10 min	After Full Body A or running
Upper-body static	Cool-down	8–10 min	After Full Body B
Foam rolling (post)	Cool-down	5–8 min	After any session
PNF (Hold-Relax)	Cool-down	8–12 min	After Full Body B or mobility day
Diaphragmatic breathing	Cool-down	2–3 min	After every session
Box breathing	Cool-down	2–4 min	After high-RPE sessions
Savasana / PMR	Cool-down	3–5 min	After long sessions

Full Body A Session Design

Full Body A emphasizes squats, horizontal pushing, glute work, and core, using movements that can be performed at home with minimal equipment. Exercises collectively target quads, glutes/hamstrings, chest, shoulders, back, calves, and core.

Perform 2–3 sets per exercise, resting 60–90 s:

- Squat (bodyweight → backpack goblet squat) — 8–15 reps
- Push-up (incline, standard, or decline) — 6–12 reps
- Hip thrust or glute bridge — 10–15 reps
- Horizontal row (inverted row, band row, or dumbbell row) — 8–12 reps

- Calf raise — 12–20 reps
- Core (plank or dead bug) — 20–40 s or 8–12 reps

Full Body B Session Design

Full Body B emphasizes unilateral lower-body work, hip hinge, pulling, and shoulder pressing, complementing Full Body A.

Perform 2–3 sets per exercise:

- Bulgarian split squat or reverse lunge — 8–12 reps per leg
- Romanian deadlift or good morning (with backpack/dumbbell) — 8–12 reps
- Horizontal or vertical pull (inverted row, band row, dumbbell row) — 8–12 reps
- Overhead push (pike push-up or shoulder press) — 6–12 reps
- Single-leg glute bridge or hip thrust — 10–15 reps per leg
- Core anti-rotation / anti-extension (side plank, shoulder taps) — 15–30 s per side

Progression and Volume Rules

- **Intensity:** Finish each set with approximately 1–3 reps in reserve (RIR), or an RPE of 7–9.
- **Weekly volume:** Target 6–12 hard sets per major muscle group per week.
- **Progression:** Once the top of the rep range is achievable on all sets with good form, add load, use a harder variation, or add one set.
- **Deloads:** If performance drops for more than one week, reduce sets by 25–30% for that muscle group temporarily.

Overall System Architecture (Local-First)

The system uses a local-first architecture: Flutter frontend communicates with a Python FastAPI backend and DuckDB database running on the same laptop, with all data stored locally. High-level components:

- Flutter app — UI for logging workouts, viewing analytics, and seeing coaching suggestions
- FastAPI backend — REST endpoints for sessions, analytics, and suggestions
- DuckDB database file (`workouts.duckdb`) — relational tables for exercises, sessions, and sets
- Optional local LLM service — accessed by the backend to generate natural-language explanations

Data Model Design (DuckDB)

Core Tables

exercises

- `exercise_id` (INTEGER, primary key)
- `name` (TEXT)
- `category` (TEXT: squat, hinge, push, pull, core)
- `primary_muscle_group` (TEXT: quads, glutes, chest, back, shoulders, etc.)

workout_sessions

- `session_id` (INTEGER or UUID primary key)
- `session_date` (DATE)
- `template` (TEXT: FullBodyA, FullBodyB, Run, Cycle, Mobility)
- `notes` (TEXT, optional)

session_sets

- `set_id` (INTEGER or UUID primary key)
- `session_id` (FOREIGN KEY → `workout_sessions`)
- `exercise_id` (FOREIGN KEY → `exercises`)
- `set_number` (INTEGER)
- `reps_completed` (INTEGER)
- `weight_kg` (REAL)
- `rir` (INTEGER)
- `rpe` (INTEGER)

Derived Analytics Views

- `weekly_muscle_volume` — per muscle group and week: `hard_sets` (sets where `rir` ≤ 3 OR `rpe` ≥ 7), `volume_load` (sets × reps × load)
- `exercise_progress` — per exercise: trend in estimated 1RM or average reps at a given load

FastAPI Backend Endpoints

POST /sessions

- Inserts a workout session and all associated sets into DuckDB.

GET /analytics/weekly

- Returns primary_muscle_group, hard_sets, and volume_load for the last 7 days.

GET /suggestions/next

- Applies rule-based logic to weekly analytics and returns:
 - suggestions — array of human-readable coaching suggestions
 - recommended_template — next session type
 - summary — short description of recent training balance

Suggestion rules:

- hard_sets < 6 and average RPE $\leq 7 \rightarrow$ add a set
- hard_sets > 15 and high RPE or fatigue noted $\rightarrow$ reduce sets
- Volume load for an exercise stalled over 3+ weeks and RPE is moderate $\rightarrow$ suggest load increase or variation change

Flutter Frontend Structure

Home / Today Screen

- Recommended workout template from /suggestions/next
- Weekly hard sets per muscle group in summary cards

Workout Logger Screen

- Select Full Body A or B (or a cardio / mobility template)
- Add sets with reps, weight, and RIR/RPE
- Submit to POST /sessions

History / Analytics Screen

- Weekly volume and hard-set charts
- Estimated 1RM progression per exercise

Suggestions / Coach Screen

- Recommended template and textual suggestions from /suggestions/next
- "Why?" button on each card $\rightarrow$ calls /coach/explain (LLM layer)
- "Ask Coach" chat interface $\rightarrow$ calls /coach/chat

Goal-Based Adaptive Weekly Planning

Goal Configuration

A `user_settings` table stores:

- Primary goal: strength, muscle_gain, or endurance
- Preferred training frequency (2–5 strength days per week)
- Equipment constraints

Target Volume Ranges by Goal

Goal	Hard sets / muscle / week	Load & rep range
Muscle gain	10–20 sets	Moderate load, 6–20 reps near failure
Strength	8–15 sets	Heavy load, 3–6 reps
Endurance / general fitness	6–10 sets	Light-moderate load, 15–25 reps

Auto Suggestion Logic

1. Read 2–4 weeks of history from DuckDB (hard sets, volume, RPE, fatigue notes).
2. Compare each muscle group's current volume to goal-specific landmarks (MEV, MAV, MRV).
3. Generate per-muscle adjustments: add sets if below MEV, reduce if above MRV, swap variation if stalled.
4. Assign next session template (Full Body A or B) based on which muscle groups most need attention.
5. Display results in the Coach screen with plain-language explanations, optionally enhanced by the local LLM.

Local LLM-Enhanced Coaching

- Ollama runs a local model (e.g., llama3 or gemma) and exposes a REST API at `http://localhost:11434`.
- FastAPI endpoints `/coach/explain` and `/coach/chat` construct structured prompts from recent analytics and rule-based suggestions and pass them to the LLM.
- The Flutter Coach screen adds a "Why?" button per suggestion (calls `/coach/explain`) and an "Ask Coach" chat interface (calls `/coach/chat`).
- Core programming logic remains deterministic in the backend; the LLM only generates explanations and conversational responses.

Data Import and External Sources

- **Strength history:** CSV/JSON import via Python ETL or `/import/csv` endpoint; exercise names resolved to `exercise_id` via mapping table.
- **Cardio / wearable data:** FIT/GPX exports (e.g., from Suunto) imported into a `cardio_sessions` table with duration, distance, and heart rate, enabling recovery-aware rules (e.g., reduce leg volume after demanding runs).
- Imported data is indistinguishable from live-logged sessions at the query level.

Implementation Roadmap

Phase 1 — Project Scaffold and Core Schema

1. Set up a Git repository with `/backend`, `/frontend`, and `/docs` directories.
2. Initialize a Python virtual environment and install FastAPI, Uvicorn, and DuckDB.
3. Create `workouts.duckdb` with SQL initialization scripts for all core tables.
4. Seed the `exercises` table with Full Body A/B exercise catalog, including warm-up and cool-down exercise types tagged by category.

Phase 2 — Backend API for Logging and Analytics

5. Implement `POST /sessions`, `GET /analytics/weekly`, and `GET /suggestions/next`.
6. Add input validation, basic error handling, and unit tests for the analytics and suggestion logic.
7. Verify endpoints manually with curl or FastAPI's built-in Swagger UI.

Phase 3 — Flutter Frontend MVP

8. Create the Flutter project with a bottom navigation structure: Home, Log, Progress, Coach.
9. Implement an `ApiClient` service pointing to the FastAPI backend.
10. Build the Workout Logger screen with exercise list, set-logging form, and session submission.
11. Build a basic Coach screen displaying the recommended template and suggestion cards.

Phase 4 — Warm-up & Cool-down Module

12. Add a `warmup_cooldown_protocols` table seeded with all protocol types, steps, durations, and session tags.
13. Implement `GET /protocols?template=FullBodyA` to return the appropriate warm-up and cool-down sequence for the selected session.
14. Build a Warm-up screen in Flutter that steps the user through each exercise with timing cues, activated before the main workout logger.
15. Build a Cool-down screen similarly, activated after the session is submitted.

16. Add activity-type detection: when the template is Run or Cycle, serve the sport-specific warm-up and cool-down protocols rather than the strength defaults.

Phase 5 — Analytics and Adaptive Planning

17. Extend backend analytics to compute volume landmarks (MEV/MAV/MRV) per muscle group over 2–4 weeks.
18. Implement goal configuration storage and the weekly planning function.
19. Enhance the Progress screen with charts for weekly hard sets, volume load trends, and estimated 1RM progression.

Phase 6 — Data Import and External Integration

20. Create import scripts or `/import/csv` endpoint for historic strength data.
21. Implement optional cardio imports (Strava exports or FIT/GPX files) into `cardio_sessions`.
22. Add an Import screen in Flutter.

Phase 7 — Local LLM Integration

23. Configure Ollama with llama3 or a small instruct model; verify the local API.
24. Implement `/coach/explain` and `/coach/chat` FastAPI endpoints.
25. Update the Coach screen with "Why?" buttons and "Ask Coach" chat interface.

Phase 8 — Polish, Testing, and Packaging

26. Add robust validation, logging, and environment-based configuration to the backend.
27. Write integration tests covering the full flow: log session → analytics → suggestions → LLM explanations.
28. Improve Flutter UI with consistent theming, fluid transitions, and mobile-optimized layouts.
29. Optionally containerize the backend with Docker for consistent local startup via `docker-compose`.

Example Logged Session

```
Date:      2026-06-27
Template:   Full Body A
Warm-up:    Dynamic full-body (8 min) + Lower-body focus add-on (5 min) + Foam rolling

Exercises:
  Goblet squat      3 × 10   15 kg backpack   RIR 2
  Incline push-up   3 × 12   bodyweight        RIR 1
  Hip thrust        3 × 15   20 kg backpack    RIR 2
  Inverted row       3 × 10   bodyweight         RIR 3
  Calf raise        2 × 20   bodyweight         RIR 1
```

Plank

3 × 30 s

RPE 8

Cool-down: Recovery walk (5 min) + Lower-body static (9 min) + Diaphragmatic breathing

From this single session, the backend computes weekly sets per muscle group, volume load per exercise, and average RPE — feeding directly into the next suggestion cycle.